

Projekt 4N: Northwest Niedersachsen Nachhaltig Neu
TV-7 Landwirtschaftlicher Strukturwandel und Klimawandel-Anpassung

Photovoltaics - light and shadow

Internal Report, Technical Report

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1 Abstract

Solar panels transform sunshine into electric power. The shaded areas around the panels receive less sunlight, which may have an impact on biodiversity. The effect of the shaded area of a row of four perpendicular solar panels is simulated. As a result, the radiation distribution is obtained and the effect of the shadow is quantified. The geometrical extent of the shaded areas are in the range of 2.5 m to the west and east side and 3.0 m to the north of the solar panels position. Minimum radiation values are observed in the three zones between the four panels, where the shadow areas receive only 50-60% of the maximum radiation, which should have a significant influence on plant growth and preferential species (like shade plants). This effect could be observed for the growth of reed grass during April and May, which is described in the last part of this document.

2 Introduction

Field margins and wildflower strips are important components for maintaining biodiversity in rural and partly urban areas. One of the tasks within the 4N project is to explore the option to combine land utilization to serve two needs at once: eco-friendly solar energy generation with photovoltaic panels and the function as biotope. A schematic view of the setup at campus Emden is shown in figure 1. Here, the setup of inclined panels facing south and also perpendicular panels facing east is combined with flowering strips. At the north side of the area hedge is marking the border of the area.

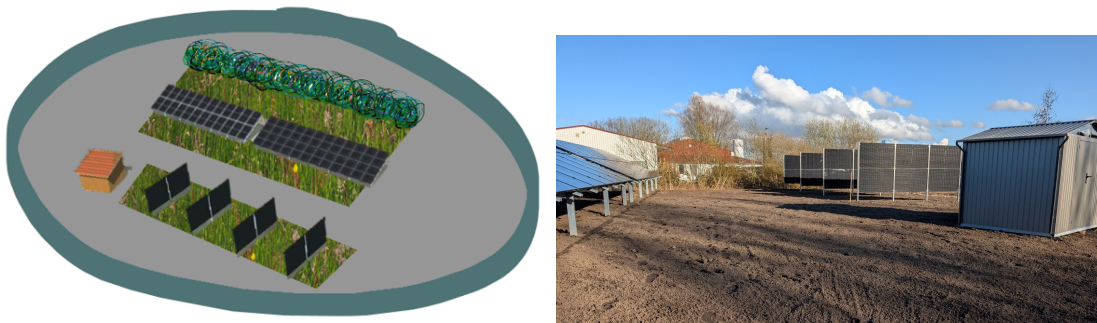


Figure 1: Left: Schematic view of the photovoltaic site at campus Emden. Right: Picture of the setup after construction in March 2025. [1]

Figure 1 (Right) also shows the presence of shadows, which are cast by the solar panels. The main task of this work is to answer the questions: Are there significant 'shadow zones'? What is the difference of light and shadow for the flowers at the ground?

The approach to this task is to calculate the shadow based on the size of the solar panels and the position of the sun. This is combined with radiation data from PVGIS. The PVGIS website [2] is provided by the European Commission and offers free and open access to meteorological data (radiation data) for photovoltaic system performance calculation. The result of the calculation is a figure which shows the radiation distribution of an area around the solar panels. The period can be chosen freely (day, week, month) as long as it is covered by the PVGIS data.

Required files

- *Timeseries_shadow_2023.csv* – contains radiation data from the PVGIS website [2] for the location at campus Emden.
- *module-single.py* – Python program to simulate the shadow around one perpendicular solar panel (for testing and separate evaluation).
- *module-multi.py* – Python program to simulate the shadow around a row of four perpendicular solar panels (shows the overlap of shadow zones).

3 Modeling procedure

The movement of the sun

The apparent movement of the sun across the sky starts with sunrise in the east and ends with sunset in the west. Figure 2 illustrates this movement. In order to calculate the shadow which is cast by an object, two angles are needed. The angle α in figure 2 describes the height of the sun above the horizon. This value is obtained from satellite data within this simulation. The angle β arises from the earth's rotation. Thus, it changes by 15° per hour, reaching the full circle with every 24 hours [3, 4]. The position of the sunrise in figure 2 indicates, that the sun does not rise every day exactly in the east. This fact is a result of the tilted axis of the earth. Therefore, the time dependency of angle β needs to be corrected for this effect, which is called 'declination' [3].

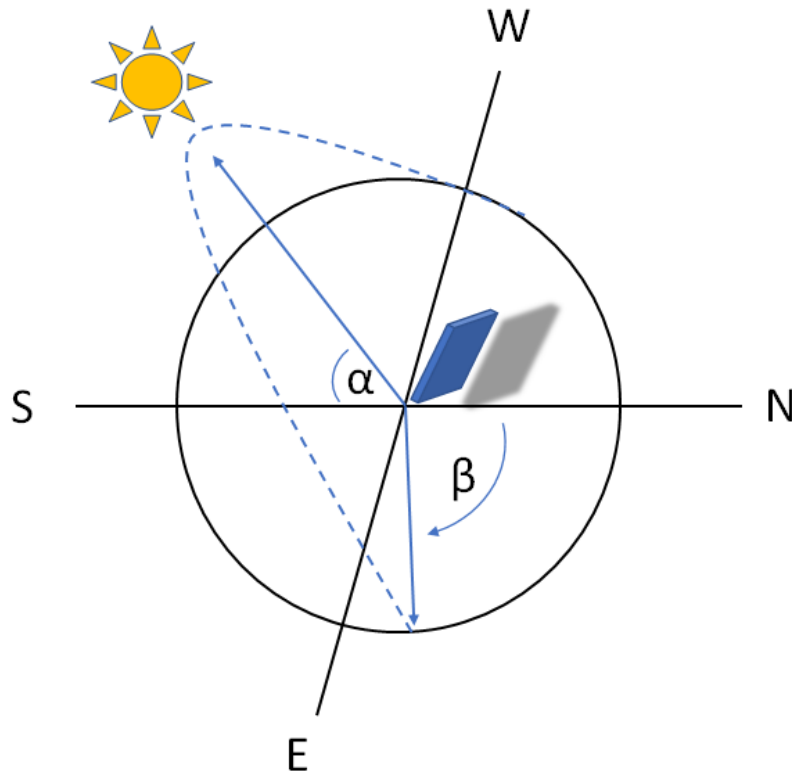


Figure 2: Apparent movement of the sun

In order to calculate the position of the shadow, a projection of the solar panel surface to the ground is generated. For each corner of the surface two parameters need to be obtained. The length of the shadow x can be calculated from the height of the sun as shown in figure 3 (Left). The shift of the shadow y from west to east is calculated from the angle which results from the earth's rotation as shown in figure 3 (Right).

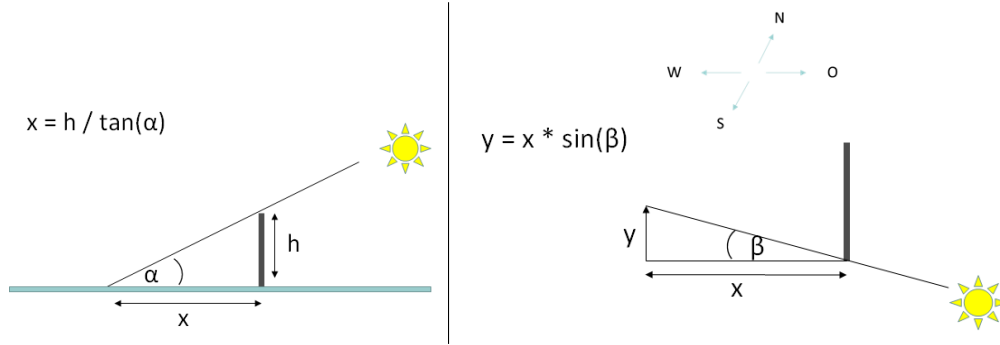


Figure 3: Left: Length of the shadow. Right: Shift from west to east.

The declination, i.e. correction for the tilted axis of the earth, is calculated from equation 1 [3]. Here, n is the number of the day within one year.

$$\Delta\beta = 23.45^\circ * \sin\left(\frac{360}{365} * (284 + n)\right) \quad (1)$$

The procedure described above is still an approximation, which is sufficient for the purpose of calculating radiation intensities on the ground. A more exact time dependency of the sun's movement can be obtained by using online tools [5, 6].

Overall program structure

This simulation tool uses radiation data from the PVGIS website [2]. This data must be obtained for the desired location. Once the respective data is downloaded, the following algorithm is applied.

1. Loop over the radiation data. Daytime is defined as being the time where the diffusive radiation is not zero.
2. Extract data for each hour: diffusive radiation, direct radiation, height of sun (angle).
3. Calculate the shadow area.
4. Loop over the entire area and assign radiation values. Inside the shadow zone only diffusive radiation is assigned, outside diffusive and direct radiation is assigned.

The overall structure is the same for the program *module-single.py* and *module-multi.py*. It contains four sections.

1. Variables declaration
2. Function declaration (4 functions)
3. Main part: perform calculation
4. Figure generation and output.

1. Variables declaration

Here, the following declarations can be found.

- File name for output graphics.
- Geometry of solar panel.
- Internal variables
- Set up and read in of PVGIS file.
- Definition of the time frame for the data in the PVGIS file and choice of simulation time frame. **Important:** Simulation time frame always needs to start at 00:00 hours! This is the starting point for the angle calculation. At 00:00 hours the sun is exactly in the north (without declination) and the angle of rotation starts at 0° .

2. Function declaration

Here, the following functions are defined.

- `make_shape(px, py)`: Creates a poly surface based on 4 points (x-y coordinates). This creates an object which represents the shadow area. The function returns this object.
- `make_modules(a)`: Creates a poly surface with offset a in x-direction. This is required to display the solar panel in the final figure.
- `beleucht(mx, my, mz, myshade, mdiff, mdir)`: Run through the 2D lattice and assign radiation intensity. The values are added and the array contains the sum over the period. At this point, there is actually only a 1D array mz , where the results are stored. The arrays mx and my provide the position.
- `deklinat(myday)`: Calculate the angle deviation for sunrise for a given day of the year.

3. Main part

Here, the three 1D arrays are initialized. mz contains results, mx and my provide the position. The size of the target area and the lattice resolution can be set here. The main calculation runs as a single loop over the entire simulation period. The loop uses two variables: i is used to access the entry in the input data array (PVGIS data) / h is used to count the hours of the day and is reset after 23 hours (in order to calculate earth rotation over one day). Finally, a 2D array mzz is created. This step is required for the plotting function.

4. Figure generation and output

The result figure is generated by a function that plots 2D fields. The x-y coordinates correspond to the number of entries in the 2D array. Therefore, the x-y axis needs to be 'corrected' in order to give numbers in meters (rather than in lattice resolution).

4 Results

The general setup of the four solar panels is given in table 1. The radiation data from the website PVGIS [2] provides radiation data as well as the height of the sun for the location of Emden. For the simulation the data was chosen for the year 2023. Figure 4 shows a top down view of one solar panel, where the shadow is calculated from sunrise to 9:00 a.m.. The time resolution is one hour, therefore three overlapping shadow areas can be observed. The values for the radiation on the ground are summed up in order to obtain the radiation distribution over the area.

Table 1: Setup of the four solar panels.

panel width	5.60 m
panel height	2.25 m
base height	0.55 m
distance between panels	5.00 m

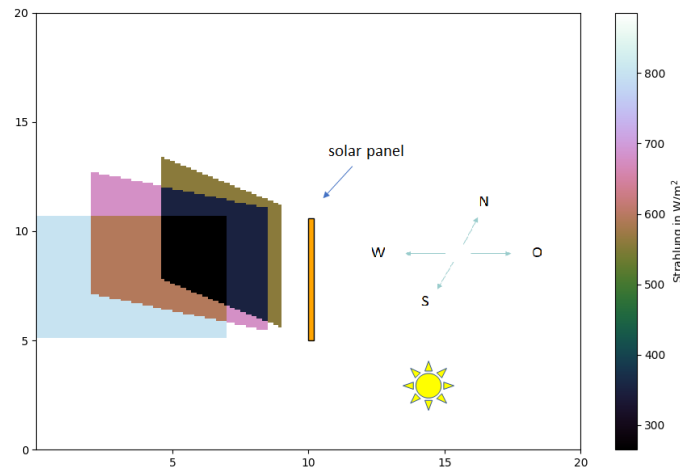


Figure 4: Top view of one panel with cast shadow until 9:00 a.m. for April 5, 2023.

The combination of four solar panels leads to an overlap of the shadow areas between the panels, as shown in figure 5. Here, the dark areas between the solar panels receive about 24% of the radiation compared to the 'bright' areas. This is an extreme situation, because it was a very sunny day with high direct radiation. Furthermore, the sun is still low above the horizon, which causes long shadows (typical for springtime).

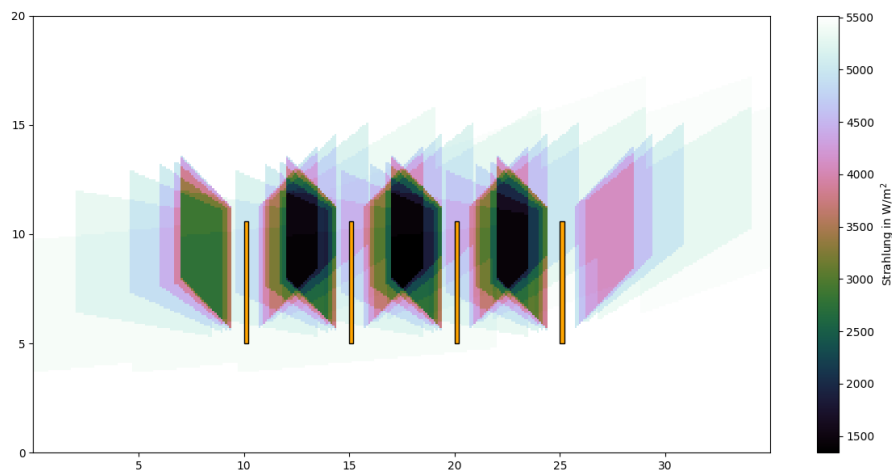


Figure 5: Top view of four panels with summarized radiation for the entire day of April 5, 2023.

The radiation data is summarized for each month, starting with April and ending with September, in order to yield an overview over the growing season. Figure 6 presents the monthly radiation data for the plants on the ground. Overall, it can be seen that (on average) the minimum values reach about 50% of the maximum radiation. These minimum values are located in the areas between the solar panels, which suffer from shadow in the morning and in the afternoon.

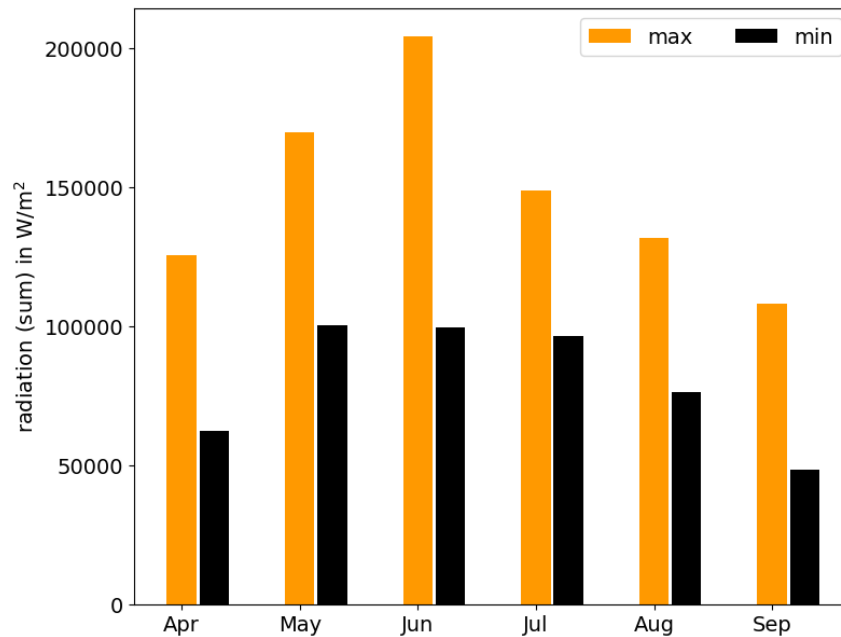


Figure 6: Monthly sums of radiation on the ground, where min values represent areas with deep shadow and max values are obtained from areas with bright sunshine.

The following figures show the radiation distribution for the separate months. A general observation for this specific location is, that the shadow during the first half of the day has a stronger impact than the shadow in the afternoon. For a single (separate) panel the shadow areas are exposed to about 60-70% of the maximum radiation. It can be concluded that a single panel may have not a strong impact on the growth of plants, since other factors besides the total radiation exposure may be important for the vegetation. The combination of four solar panels lead to a decrease of radiation inside the shadow areas. Here, the shadow areas receive only 50-60% of the maximum radiation, which should have a significant influence on plant growth and preferential species (like shade plants). The geometrical extent of the shade areas can be extracted from figures 7 to 12. Major shading effects are observed in the range of 2.5 m to the west and east side and 3.0 m to the north. Minimum radiation values are observed in the three zones between the four panels.

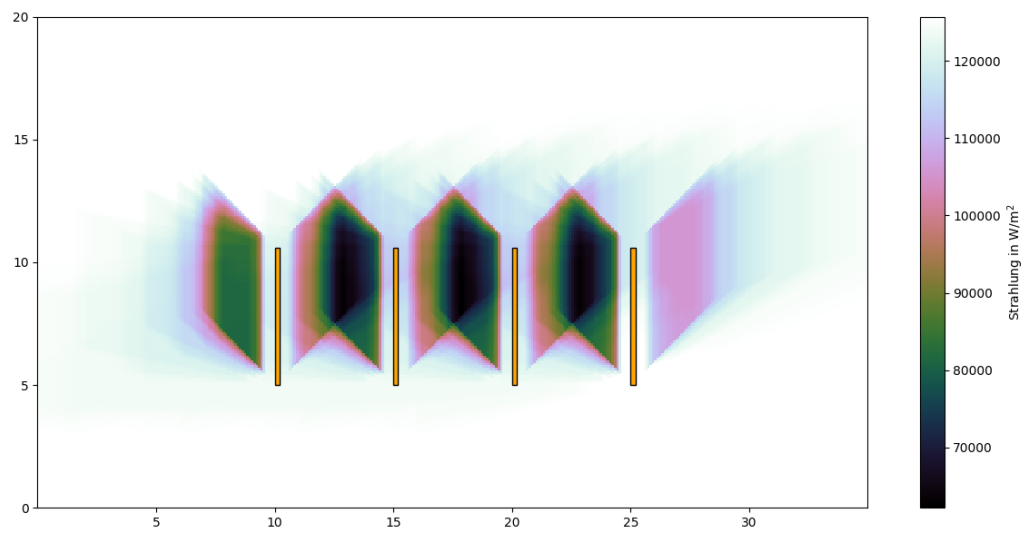


Figure 7: Radiation distribution around four panels in April 2023.

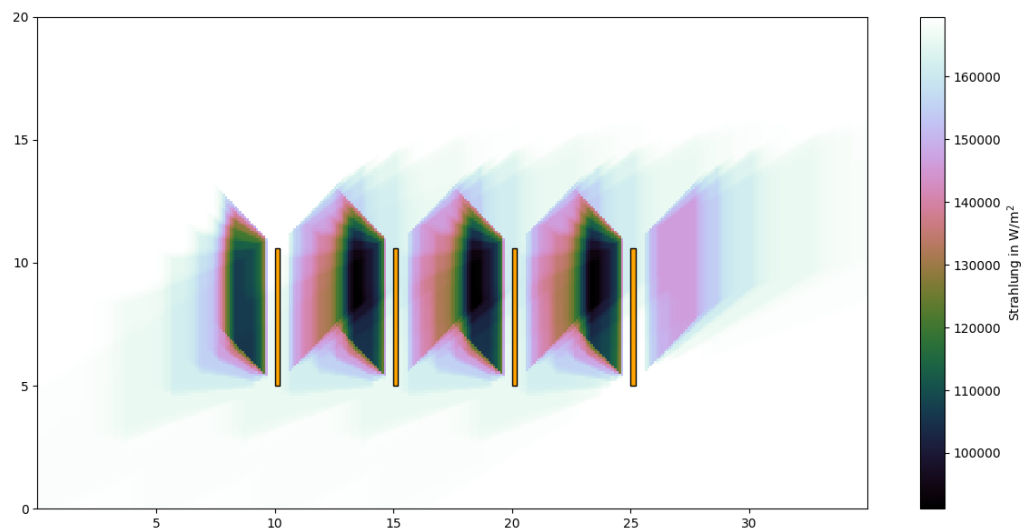


Figure 8: Radiation distribution around four panels in May 2023.

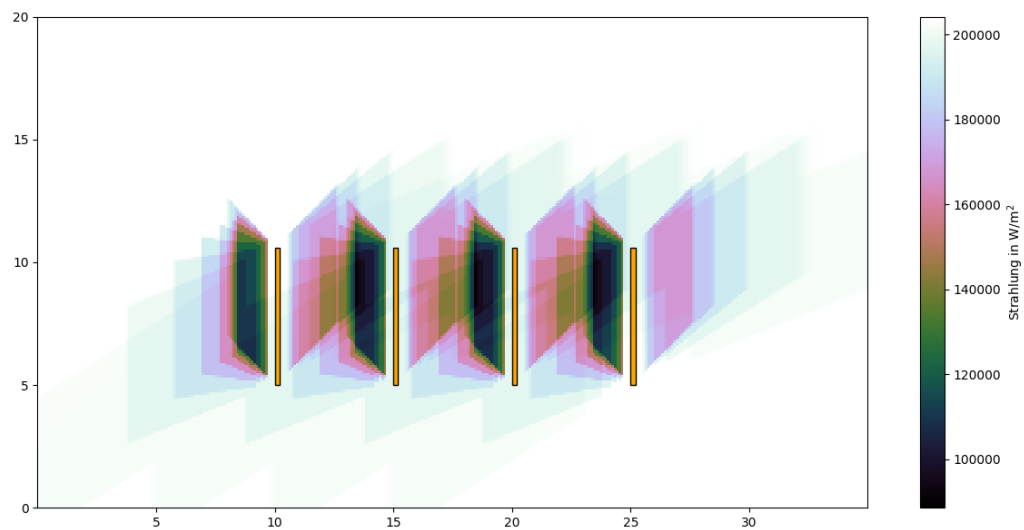


Figure 9: Radiation distribution around four panels in June 2023.

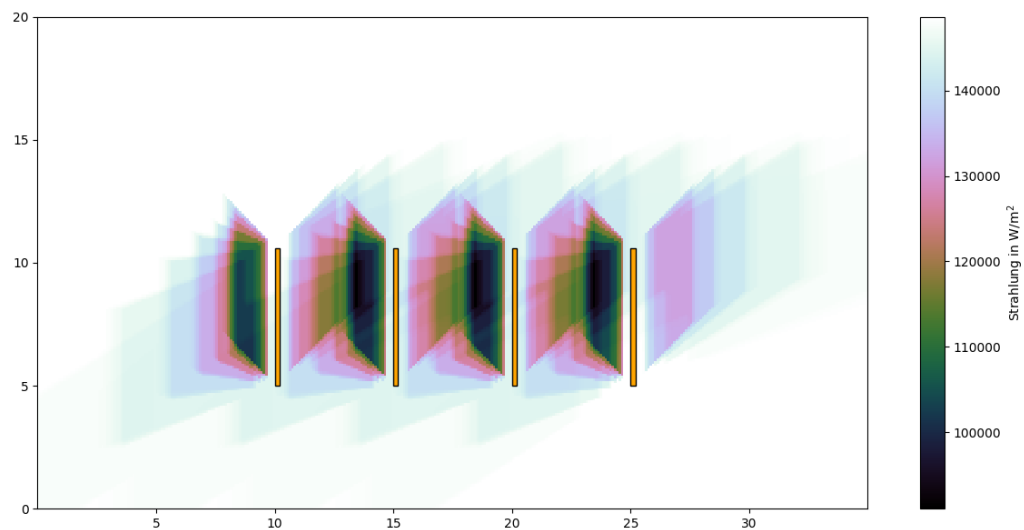


Figure 10: Radiation distribution around four panels in July 2023.

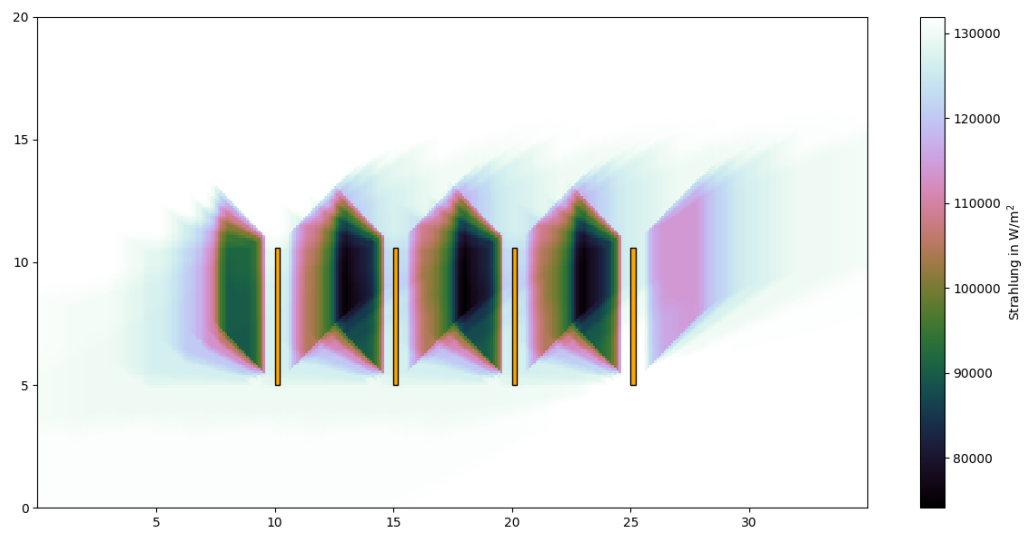


Figure 11: Radiation distribution around four panels in August 2023.

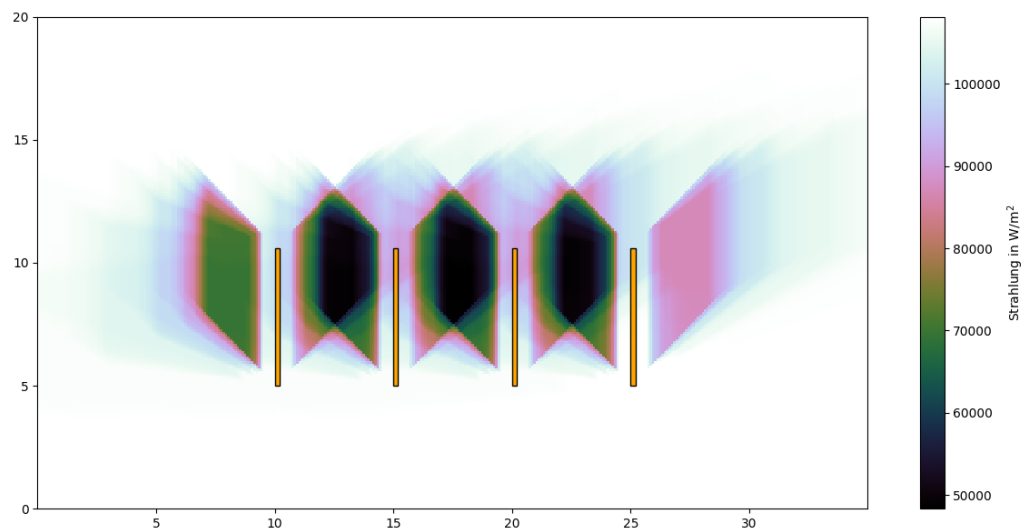


Figure 12: Radiation distribution around four panels in September 2023.

5 Validation / reality check

The simulated data within this work depends on several model assumptions. It is always helpful to check whether the results correspond to reality. The first attempt was to take photos from the cast shadow and compare them with the simulations. Unfortunately, it turned out that the photos taken by smartphone can not be processed straight away. Due to the effect of the small lense and the perspective view, the in depth information is modified in a way that straight measurements are not possible. Therefore, two other methods are presented.

Rectangular shadow

Figure 13 (Left) shows a photo taken on May 22, 2025. At 08:40 hours the sun was more or less exactly in the east and the solar module casted a perfect rectangular shadow. Figure 13 (Right) shows the simulation data for this time. Since the time resolution is one hour, the data for 08:00 and 09:00 hours is shown (corrected for summer time). It can be seen that the perfect rectangular shape falls exactly between these two hours. It should be noted that the simulated data is actually for May 21, 2023. The reason for this lies in the underlying satellite data. For May 22 in 2023 was a cloudy day without direct radiation.

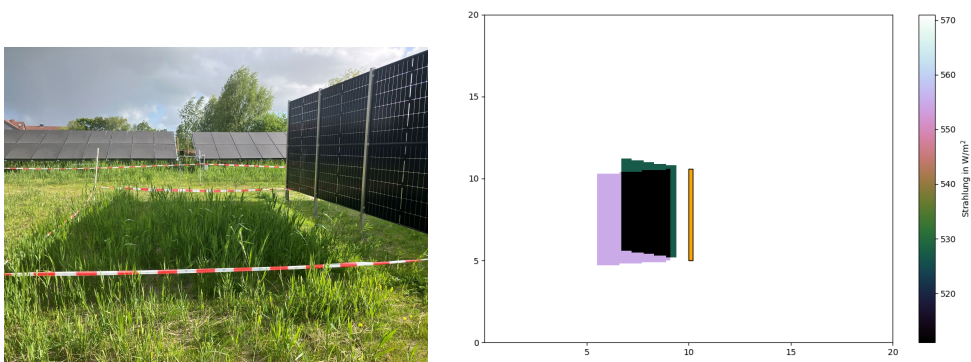


Figure 13: Left: Rectangular shadow on May 22 at 08:40 hours. Right: Simulation for this day at 08:00 and 09:00 hours.

Effect on plant growth

The main purpose of this simulation tool is to provide radiation data for the plants that grow between the solar panels. Light is an important resource for plants and the distribution of shadow areas should have an impact on growth. Figure 14 shows a photo from the west side of the solar panels. Here, the reed grass grows within an area of a triangle zone.

This triangle zone corresponds reasonably well with the area, which receives the most light – as shown in figure 15 for the months of April and May. The border line of the reed grass follows more or less the line, where the shadow has the strongest impact. For April the shadow area receives about 65% of the total radiation and for May the value is about 63%. Besides the effect of light (for photosynthesis) also the effect of warmth may be important for the plants to grow.



Figure 14: Reed grass growing at the west side of the panels (on May 21).

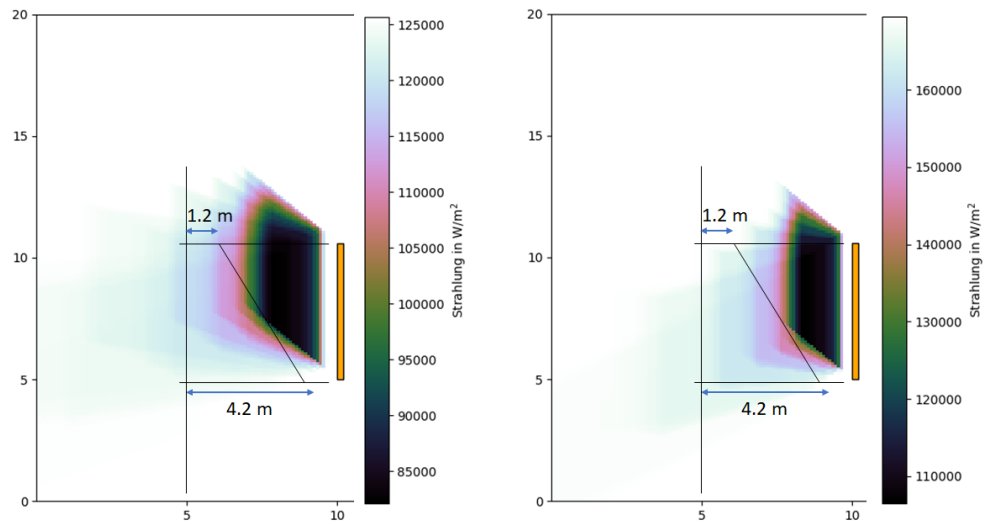


Figure 15: Simulation data for the month of April (Left) and May (Right).

References

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